🇬🇧 GeoAstro — Statistical Module

User Guide — Quick Start

This guide describes the practical workflow for using the GeoAstro statistical module: preparing data, running permutation analysis, exporting results, and performing H/F comparisons.

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1) Prepare the CSV file

Use one of the provided templates (FR or EN).

Each row corresponds to one individual.

Required fields:

- Name / Identifier
- Date of birth
- UT time
- Latitude
- Longitude

⚠️ Field formats must be respected to ensure compatibility with the statistical module and the graphical tools.

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2) Import the file and run the analysis

Open the **Analysis** tab.

Procedure:

1. Load the CSV file
2. Select display language
3. Choose the number of permutations (1,000 or 10,000)
4. Click **Run statistical analysis**

When processing is complete, the results table is displayed.

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3) About the results

The table includes:

- observed values,
- permutation-based means,
- z-scores,
- empirical probabilities (over-valuation / under-valuation).

⚠️ Results indicate deviations from random expectation.

They are not interpretive in themselves and must be understood within the methodological framework described in the GeoAstro protocol.

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4) Export results

Click **Export results (CSV)**.

The exported file is formatted for direct use in:

- the graphical module (histograms)
- the curve module (Gaussian / KDE)

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5) H/F comparison (optional)

Open the **H/F Comparison** tab.

Procedure:

1. Load a CSV file composed of “Male” cases
2. Load a CSV file composed of “Female” cases
3. Generate the comparison table
4. Export the dedicated CSV file

This file can then be used in the H/F graphical module (**Histograms** and **Curves** tabs).

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6) Generate Charts — Histograms

Open the **Histograms** tab.

This module is used to visualise the results as comparative histograms, based on a CSV file exported from the statistical analysis.

Procedure:

- Load the results CSV file
- Select the population:
  - Global
  - Male / Female[1]
- Select the category to display (planets, RET families, zodiac signs, etc.)
- Optional:
  - choose whether or not to display z-scores and p-values
- Click **Generate chart**

The chart can be exported as a PNG file using the **Export PNG** button.

⚠ Histograms show deviations relative to permutation-based averages. They do not constitute an interpretation in themselves, but provide a visual support for analysis.

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7) Generate Charts — Curves (Gauss / KDE)

Open the **Curves** tab.

This module is used to visualise the statistical distribution of an element in the form of:

- a Gaussian curve (normal distribution), or
- a KDE curve (empirical smoothing)

Procedure:

- Load the results CSV file
- Select the population (Global; Male / Female[1])
- Select the category and the element to display
- Select the curve type:
  - **Gaussian**
  - **KDE**[2] (if available)
- Click **Generate chart**

The chart can be exported as a PNG file using **Export as PNG**.

⚠ The curve shows the position of the observed score relative to the distribution obtained from permutations.

It provides visual support for analysis and should not be interpreted in isolation.

[1] CSV generated by the Male/Female Comparison tab.

[2] KDE curve generation is only possible if, during statistical analysis (Analysis tab), the "Include KDE file" option was selected before starting the calculation.
Otherwise, only Gaussian curves are available.

8) Recommended practices

- use homogeneous and well-documented samples,
- avoid mixed datasets for H/F comparison,
- keep source datasets and exported outputs together.

📎 Full documentation

For methodological details, statistical justification and worked examples:
→ refer to the **GeoAstro Operating Protocol**.